

Transformation with E. Coli.

Introduction

After plasmid cloning, one can use E. Coli. to create colonies and amplify our product

Based on this [NEB protocol](#).

Materials

- > NEB Stable Competent *E. coli* cells
 - > freezer...
- > Plasmid DNA
 - > From plasmid cloning protocol
- > NEB 10-beta/Stable Outgrowth Medium
 - > fridge, in FIA box
- > Selection plates
 - > fridge
- > LB broth
 - > fridge!
- > Ice

Procedure

Transformation

1. Thaw a tube of NEB Stable Competent *E. coli* cells on ice for 10 minutes.

They are already aliquoted.

2. Add 1-2 μ l containing 100 pg – 100 ng of plasmid DNA to the cell mixture. Carefully flick the tube 4-5 times to mix cells and DNA. Do not vortex.

I did 2 μ L

3. Place the mixture on ice for 30 minutes. Do not mix.

4. Heat shock at exactly 42°C for exactly 30 seconds. Do not mix. Place on ice for 5 minutes. Do not mix.

5. Pipette 950 μ l of room temperature NEB 10-beta/Stable Outgrowth Medium into the mixture and place at 30°C for 60 minutes. Shake the tube horizontally at 250 rpm or rotate.

put in the shakey incubator thing in bacteria room

6. Warm selection plates (do 2 per condition) to 30°C (in incubator).

7. Mix the cells thoroughly by flicking the tube and inverting.
8. Create dilutions (1:1, 1:10, 1:100, 1:1000 (do two per selection plate). Then spread 50-100 µl of cells or diluted cells onto a selection plate.

we did 50 uL
You want to do dilutions in order to be able to isolate singular colonies
Use dull rake thing to spread cells; be careful to not pass "boundary"
9. Incubate plates 24 hrs at 30°C or overnight at 37°C.

Colonies

10. You should see opaque colonies that lessen as the dilution ratio increases. If they are too transparent, leave to grow a little longer.
11. Add 5 mL of LB broth to 15ml falcon tube. Do one tube per colony.

i think this is the point where you can and should make bacterial glycerol stocks
12. Isolate >3 colonies with the circle tool and place them in the broth-filled tubes.
13. Incubate overnight in the bacteria cage at 37C
14. Run [QIAprep Spin Miniprep Kit](#)
15. Run Qubit
16. Send to [eurofins](#) for WPS. Find materials and instructions in top drawer in bacterial lab. (want 30 ng in 20 uL)

send before 15:00!